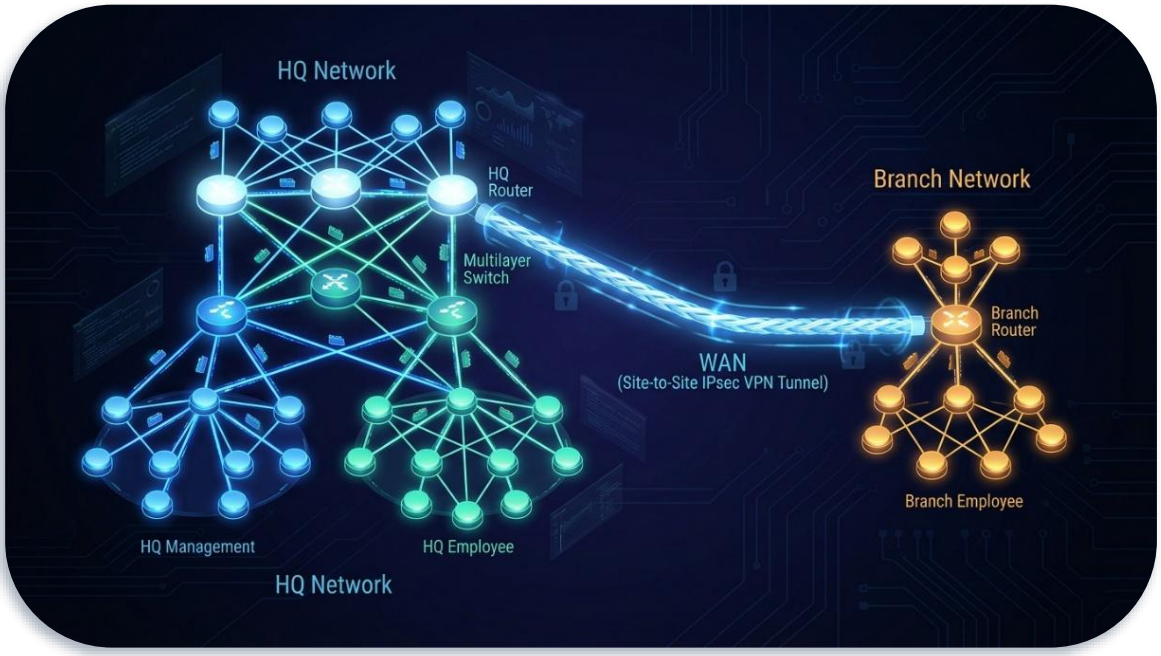
Enterprise Dual-Branch Network Design Documentation



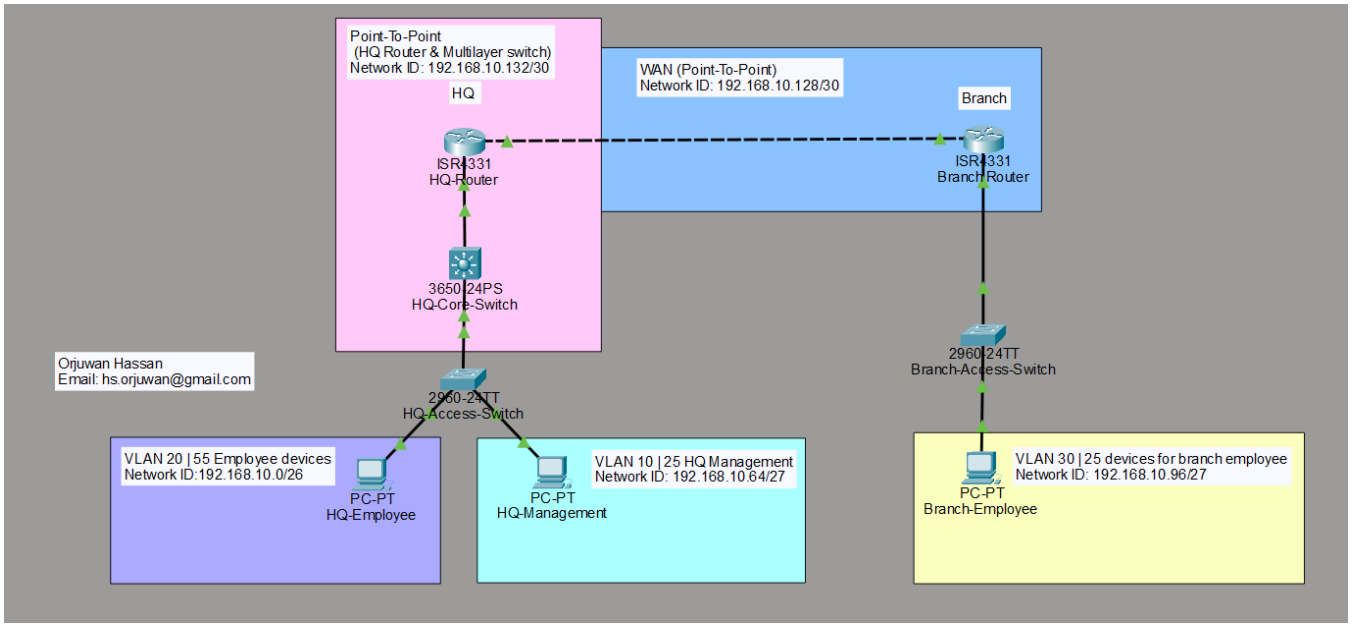
Designed & documented by: Orjuwan Hassan

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1. Network Description

Designing and developing the network infrastructure for an organization consisting of an HQ and a branch, managing data flow, ensuring security, and encrypting the connection between them across a WAN.

2. Network Topology



3. Hardware Components

HQ: Router (ISR4331), Multilayer Switch (3650 – 24PS), Access Switch (2960 – 24TT), 25 devices for HQ management employee (PC-PT), 55 devices for HQ employee (PC-PT)

Branch: Router (ISR4331), Access Switch (2960 – 24TT), 25 devices for branch employee (PC-PT)

4. Network Address: 192.168.10.0/24

4.1 VLSM Addressing Table

Broadcast	Last Usable IP	First Usable IP	Subnet Mask	Prefix &Network ID	Devices	VLAN
192.168.10.63	192.168.10.62	192.168.10.1	255.255.255.192	192.168.10.0/26	55	**HQ Employee (VLAN 20)**
192.168.10.95	192.168.10.94	192.168.10.65	255.255.255.224	192.168.10.64/27	25	**HQ Management (VLAN 10)**
192.168.10.127	192.168.10.126	192.168.10.97	255.255.255.224	192.168.10.96/27	25	**Branch Employee (VLAN 30)**
192.168.10.131	192.168.10.130 (R2)	192.168.10.129 (R1)	255.255.255.252	192.168.10.128/30	2	**Routers Link WAN Point-) (to-Point**
192.168.10.135	192.168.10.134/30 (Multilayer Switch)	192.168.10.133/30 (HQ Router)	255.255.255.252	192.168.10.132/30	2	**HQ Router & Multilayer switch Link (Point-To-Point)**

5. Configuration steps

A. Multilayer Switch 3650

1. "**ip routing**" command to enable routing between VLANs
2. Creating VLANs and SVIs:
    - Create VLAN 10 & VLAN 20
    - Configure addresses to SVIs: interface vlan 10 (192.168.10.65) & interface vlan 20 (192.168.10.1)
3. Configure DHCP pools:
    - Exclude the first usable addresses of the two VLANs by using the command:

“ip dhcp excluded-address 192.168.10.1 & 192.168.10.65”

- Create the following pools: MGT_POOL & EMP_POOL

4. OSPF Routing:
   Configure the employee & management network and the router link as “**router ospf 1**” in “**area 0**”

B. HQ & Branch Routers

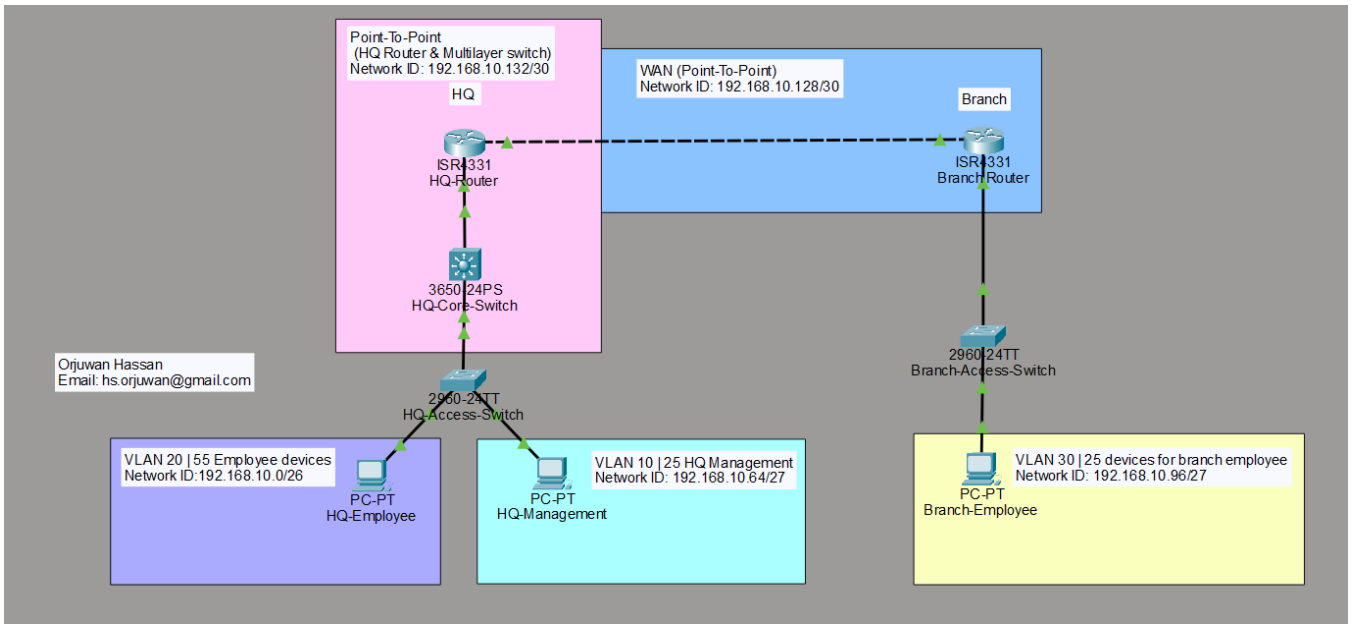
1. **Interfaces configurations**: Configure WAN and LAN interfaces with the appropriate IP addresses and ensure they are activated using the no shutdown command.
2. **Enable OSPF:** Create neighborhood and exchanging routing information between the HQ and branch.
3. Apply security with **SSHv2** protocol:
   - Specify domain name “**company.local**”
   - Generate RSA keys with 1024 bits
   - Adjust the user and their privilege “**privilege 15 secret**”
   - Lock vty lines with the command “**transport input ssh**”

C. Site-to-Site IPsec VPN

- IKE Phase 1: create the “isakmp policy 10” policy by using AES encryption, SHA, and the shared key.
- IPsec Phase 2: adjust “transform-set MY-SET esp-aes 128 esp-sha-hmac”
- ACL 101
- Apply crypto map: link policies and apply the map on WAN interfaces.

6. verification & Testing

A. Network Topology



B. Routing Table

```
HQ-Router(config)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.10.0/24 is variably subnetted, 7 subnets, 4 masks
O       192.168.10.0/26 [110/2] via 192.168.10.134, 00:08:36, GigabitEthernet0/0/1
O       192.168.10.64/27 [110/2] via 192.168.10.134, 00:08:36, GigabitEthernet0/0/1
O       192.168.10.96/27 [110/2] via 192.168.10.130, 00:08:36, GigabitEthernet0/0/0
C       192.168.10.128/30 is directly connected, GigabitEthernet0/0/0
L       192.168.10.129/32 is directly connected, GigabitEthernet0/0/0
C       192.168.10.132/30 is directly connected, GigabitEthernet0/0/1
L       192.168.10.133/32 is directly connected, GigabitEthernet0/0/1

HQ-Router(config)#
```

C. Neighbors Table

```
HQ-Router(config)#do show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.10.130    1   FULL/BDR        00:00:39    192.168.10.130  GigabitEthernet0/0/0
192.168.10.134    1   FULL/DR         00:00:39    192.168.10.134  GigabitEthernet0/0/1
HQ-Router(config)#
```

D. Connection Test Using "Ping"

Connection test between VLAN 30 (Branch-Employee) to VLAN 20 (Employee)

```
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time<1ms TTL=125
Reply from 192.168.10.2: bytes=32 time=1ms TTL=125
Reply from 192.168.10.2: bytes=32 time=10ms TTL=125
Reply from 192.168.10.2: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>
```

E. Encryption

```
HQ-Router(config)#do sh crypto ipsec sa

interface: GigabitEthernet0/0/0
    Crypto map tag: MY-MAP, local addr 192.168.10.129

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (192.168.10.0/255.255.255.128/0/0)
   remote  ident (addr/mask/prot/port): (192.168.10.96/255.255.255.224/0/0)
   current_peer 192.168.10.130 port 500
    PERMIT, flags={origin_is_acl,}
   #pkts encaps: 11, #pkts encrypt: 11, #pkts digest: 0
   #pkts decaps: 11, #pkts decrypt: 11, #pkts verify: 0
   #pkts compressed: 0, #pkts decompressed: 0
   #pkts not compressed: 0, #pkts compr. failed: 0
   #pkts not decompressed: 0, #pkts decompress failed: 0
   #send errors 0, #recv errors 0

     local crypto endpt.: 192.168.10.129, remote crypto endpt.:192.168.10.130
     path mtu 1500, ip mtu 1500, ip mtu idb GigabitEthernet0/0/0
     current outbound spi: 0x1CD8B5CE(483964366)

     inbound esp sas:
      spi: 0x408FA631(1083156017)
        transform: esp-aes 128 esp-sha-hmac ,
        in use settings ={Tunnel, }
        conn id: 2007, flow_id: FPGA:1, crypto map: MY-MAP
        sa timing: remaining key lifetime (k/sec): (4525504/2528)
        IV size: 16 bytes
        replay detection support: N
        Status: ACTIVE
```

After testing the connection multiple times (notice the increase in #pkts encaps & #pkts decaps)

```
HQ-Router(config)#do sh crypto ipsec sa

interface: GigabitEthernet0/0/0
    Crypto map tag: MY-MAP, local addr 192.168.10.129

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (192.168.10.0/255.255.255.128/0/0)
   remote  ident (addr/mask/prot/port): (192.168.10.96/255.255.255.224/0/0)
   current_peer 192.168.10.130 port 500
    PERMIT, flags={origin_is_acl,}
   #pkts encaps: 15, #pkts encrypt: 15, #pkts digest: 0
   #pkts decaps: 19, #pkts decrypt: 19, #pkts verify: 0
   #pkts compressed: 0, #pkts decompressed: 0
   #pkts not compressed: 0, #pkts compr. failed: 0
   #pkts not decompressed: 0, #pkts decompress failed: 0
   #send errors 0, #recv errors 0

     local crypto endpt.: 192.168.10.129, remote crypto endpt.:192.168.10.130
     path mtu 1500, ip mtu 1500, ip mtu idb GigabitEthernet0/0/0
     current outbound spi: 0x1CD8B5CE(483964366)

     inbound esp sas:
      spi: 0x408FA631(1083156017)
        transform: esp-aes 128 esp-sha-hmac ,
        in use settings ={Tunnel, }
        conn id: 2007, flow_id: FPGA:1, crypto map: MY-MAP
        sa timing: remaining key lifetime (k/sec): (4525504/1940)
        IV size: 16 bytes
        replay detection support: N
        Status: ACTIVE
```